

**Fourth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ANALOG CIRCUITS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
 - (2) Assume suitable data wherever necessary.
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1.
 - (a) With the help of an equivalent circuit derive output resistance R_{of} and R_{of}' for a voltage series feedback amplifier. 5(CO1,2)
 - (b) Illustrate with the help of a circuit diagram that the gain of emitter follower is unity. Also identify the feedback topology used in this circuit. 5(CO1,2)
 2.
 - (a) Prove that the efficiency of class A transformer coupled power amplifier is 50%. 5(CO3)
 - (b) Explain the reason for better performance of class C followed by class B and then class A power amplifier when evaluated on the basis of efficiency. 5(CO3)
 3. Derive the formula for frequency of oscillations and feedback factor for Hartley oscillator. 10(CO1,2)
 4. Derive voltage gain, input resistance and output resistance for a single input unbalanced output differential amplifier. 10(CO3)

5. (a) Derive gain and input resistance for OP-AMP amplifier in inverting configuration. 6(CO4)
- (b) An open loop OP-AMP amplifier in non-inverting configuration has an input of 1 mVpp. Sketch input and output waveforms. Indicate time period and voltages of input and output on it. Assume supply voltage is ± 12 V and input frequency is 1 kHz. 4(CO4)
6. (a) Design a second order HPF with high cut-off frequency of 3 kHz. 5(CO4)
- (b) Design a monostable multivibrator using timer IC 555 with an on time of 1 ms. If it is to be triggered externally, calculate typical frequency of the trigger source. 5(CO5)

